



Student number: _____

GIRRAWEE HIGH SCHOOL

2020

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided at the back of this paper
- In section II, show relevant mathematical reasoning and/or calculations

**Total marks:
70**

Section I – 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 60 marks

- Attempt all questions
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1-10

1. The unit vector in the direction of $\underline{u} = \underline{i} - 2\underline{j}$ is

- A. $5(\underline{i} - 2\underline{j})$
- B. $\frac{1}{5}(\underline{i} - 2\underline{j})$
- C. $\sqrt{5}(\underline{i} - 2\underline{j})$
- D. $\frac{1}{\sqrt{5}}(\underline{i} - 2\underline{j})$

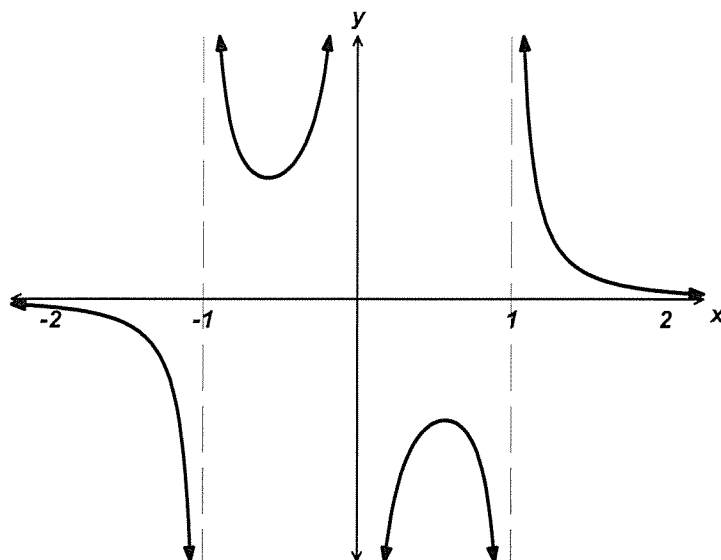
2. A vector perpendicular to $3\underline{i} + 4\underline{j}$ and with magnitude 5 is

- A. $-4\underline{i} - 3\underline{j}$
- B. $5(4\underline{i} - 3\underline{j})$
- C. $4\underline{i} - 3\underline{j}$
- D. $\sqrt{5}(4\underline{i} - 3\underline{j})$

3. Which of the following is the derivative of $\tan^{-1}(3x)$?

- A. $3 \tan^{-1} x$
- B. $\frac{3}{1 + x^2}$
- C. $\frac{3}{1 + 9x^2}$
- D. $3 \sec^2 3x$

4.



The graph above shows $y = \frac{1}{f(x)}$.

Which one of the following equations best represents $f(x)$?

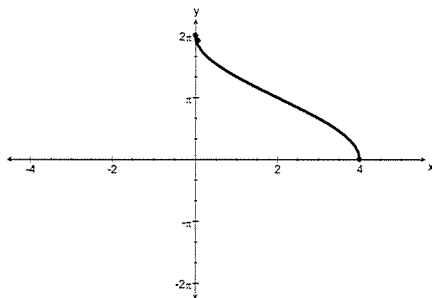
- A. $f(x) = x^2 - 1$
 - B. $f(x) = x(x^2 - 1)$
 - C. $f(x) = x^2(x^2 - 1)$
 - D. $f(x) = x^2(x^2 - 1)^2$
5. What is the value of $\sin 2x$ given that $\sin x = \frac{\sqrt{3}}{2}$ and x is obtuse?
- A. $-\frac{\sqrt{3}}{4}$
 - B. $-\frac{\sqrt{3}}{2}$
 - C. $\frac{\sqrt{3}}{4}$
 - D. $\frac{\sqrt{3}}{2}$
6. Four females and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate?
- A. $4! \times 4!$
 - B. $3! \times 4!$
 - C. $3! \times 3!$
 - D. $2 \times 3! \times 3!$

7. If $\sin(\alpha + \beta) = a$ and $\sin(\alpha - \beta) = b$, then $\sin \alpha \cos \beta$ is equal to

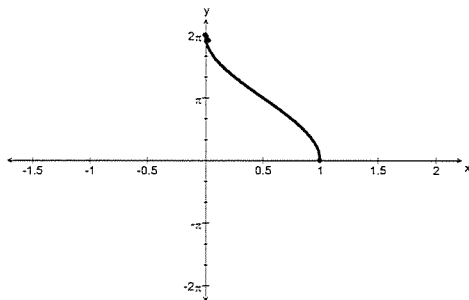
- A. $\sqrt{a^2 + b^2}$
- B. $\sqrt{ab}$
- C. $\sqrt{a^2 - b^2}$
- D. $\frac{a + b}{2}$

8. Which of the graphs below shows $y = 2 \cos^{-1}\left(\frac{x}{2} - 1\right)$?

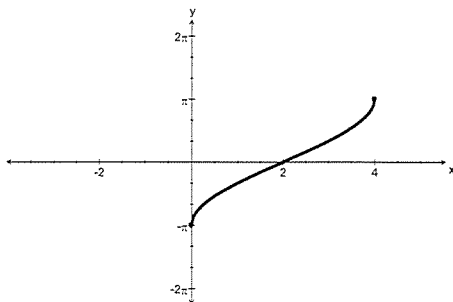
A.



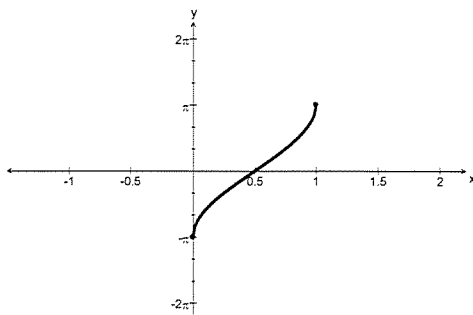
B.



C.



D.



9. $\sin 3x \sin 4y =$

A. $\frac{1}{2}[\cos(3x - 4y) - \cos(3x + 4y)]$

B. $\frac{1}{2}[\cos(3x + 4y) - \cos(3x - 4y)]$

C. $\frac{1}{2}[\cos(3x - 4y) - \cos(3x - 4y)]$

D. $\frac{1}{2}[\cos(3x - 4y) - \sin(3x + 4y)]$

10. When the polynomial $P(x)$ is divided by $(x - 2)$ and $(x + 1)$ the respective remainders are 5 and 8.

What is the remainder when $P(x)$ is divided by $(x - 2)(x + 1)$?

A. $7 - x$

B. $x - 7$

C. $x + 7$

D. $-7 - x$

Section II

60 marks

Attempt all questions

Allow about 1 hour and 45 minutes for this section

Start each question on a new page in the answer booklet provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is available on request.

Question 11 (14 marks)

a. Solve $\frac{3}{4x-1} \geq 2$ [3]

b. Differentiate $e^x \tan^{-1} x$ [2]

c. Consider the curve $f(x) = x^2 - 4x + 5$

(i) Find the largest possible positive domain for which $f(x)$ has an inverse function $f^{-1}(x)$. [1]

(ii) Find the point(s) of intersection of $y = f(x)$ and $y = f^{-1}(x)$ in the domain determined in part (i). [2]

(iii) State the domain of $y = f^{-1}(x)$. [1]

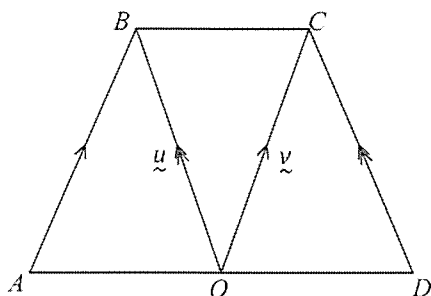
(iv) What is the equation of $y = f^{-1}(x)$? [2]

(v) For the restricted domain, sketch the function and the inverse function on the same number plane.
Clearly label the graphs and show the main features. [3]

Question 12 (11 marks)

a. Evaluate $\int_0^{15} \frac{x}{\sqrt{x+1}} dx$ using the substitution $u^2 = x + 1$. [2]

b.



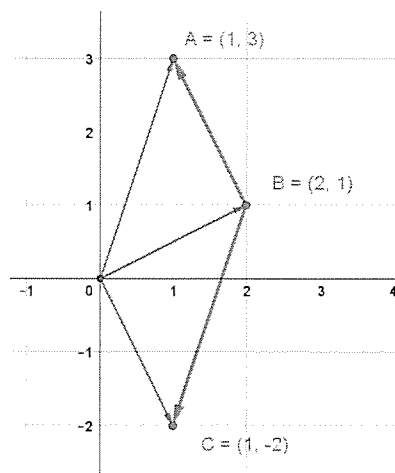
$$AB \parallel OC ; DC \parallel OB ; \overrightarrow{OB} = \underline{u} ; \overrightarrow{OC} = \underline{v}$$

$$AB = OB = OC = DC$$

(i) Express $\overrightarrow{AD}$ in terms of $\underline{u}$ and $\underline{v}$ [2]

(ii) Express $\overrightarrow{BD}$ in terms of $\underline{u}$ and $\underline{v}$ [1]

c.



A, B and C are points defined by the position vectors

$$\underline{a} = \underline{i} + 3\underline{j} , \underline{b} = 2\underline{i} + \underline{j} \text{ and } \underline{c} = \underline{i} - 2\underline{j} \text{ respectively.}$$

(i) Find $\overrightarrow{BA}$ and $\overrightarrow{BC}$ [2]

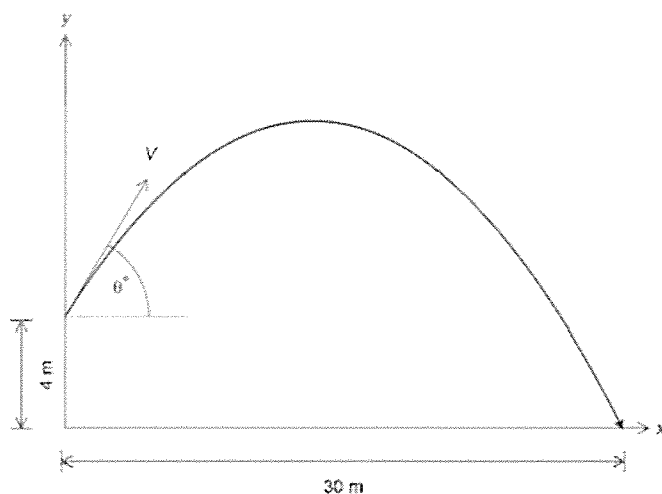
(ii) Find $|\overrightarrow{BA}|$ and $|\overrightarrow{BC}|$ [2]

(iii) Find $\angle ABC$ [2]

Question 13 (13 marks)

- a. Use mathematical induction to prove that $3^{2n+1} + 2^{n+2}$ is divisible by 7 for integers $n \geq 1$. [3]

b.



A rock is projected with a speed of $V \text{ ms}^{-1}$ from a point 4 metres above a flat ground. The angle of projection to the horizontal is θ as shown.

- (i) Taking the ground as the origin and the acceleration due to gravity as 10 ms^{-2} , show that $x = Vt \cos \theta$ and $y = Vt \sin \theta - 5t^2 + 4$ [2]

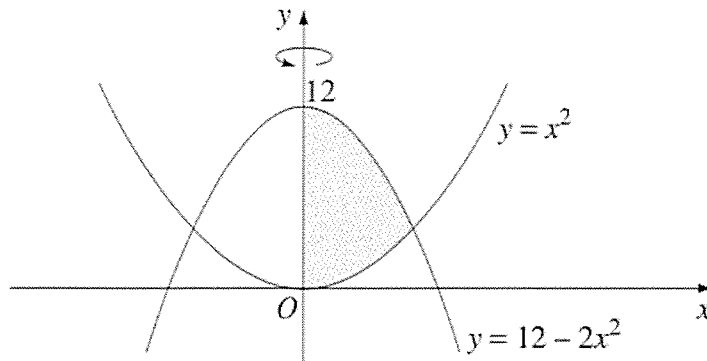
- (ii) If the rock hits the ground 30 metres away, find the value of V if $\theta = \tan^{-1}\left(\frac{5}{12}\right)$ [3]

- (iii) Find the maximum height reached. [2]

- c. Let $P(x) = x^3 + ax^2 + bx + 5$ where a and b are real numbers. Find the values of a and b given that $(x - 1)^2$ is a factor of $P(x)$. [3]

Question 14 (11 marks)

a.



The graphs of the curves $y = x^2$ and $y = 12 - 2x^2$ are shown in the diagram.

(i) Find the points of intersection of the two curves. [1]

(ii) The shaded region between the curves and the y -axis is rotated about the y -axis. By splitting the shaded region into two regions, or otherwise, find the volume of the solid formed. [3]

b. A salad, which is initially at a temperature of 25°C , is placed in a refrigerator that has a constant temperature of 3°C . The cooling rate of the salad is proportional to the difference between the temperature in the refrigerator and the temperature, T , of the salad. That is T satisfies the equation

$$\frac{dT}{dt} = -k(T - 3)$$

where t is the number of minutes after the salad is placed in the refrigerator.

(i) Show that $T = 3 + Ae^{-kt}$ satisfies this equation. [1]

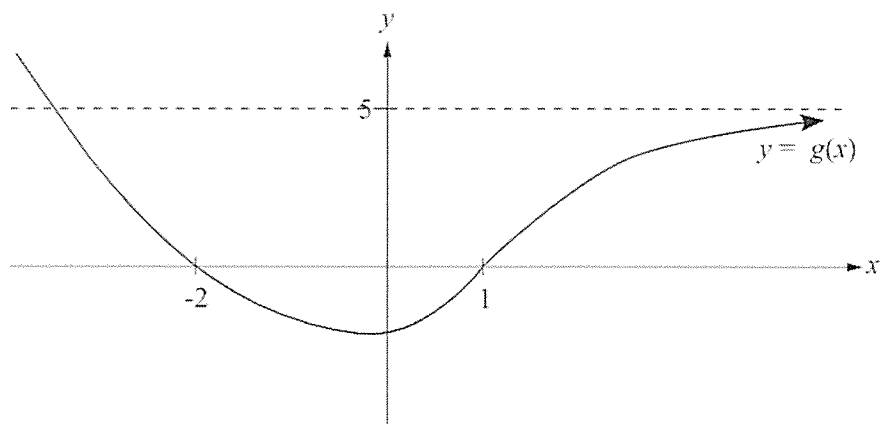
(ii) The temperature of the salad is 11°C after 10 minutes. Find the temperature of the salad after 15 minutes. [3]

c. How many numbers greater than 6000 can be formed with the digits 1, 4, 5, 7, 8 if no digit is repeated? [2]

Question 15 (11 marks)

- a. (i) Show that

$$2\sin 2x - 3\cos 2x - 3\sin x + 3 = \sin x(6\sin x + 4\cos x - 3)$$
 [2]
- (ii) Express $6\sin x + 4\cos x$ in the form $R\sin(x + \alpha)$ where
 $R > 0$ and $0 < \alpha < \frac{\pi}{2}$ [2]
- (iii) Hence, solve $2\sin 2x - 3\cos 2x - 3\sin x + 3 = 0$ for $0 \leq x < \pi$
 Answer in radians, correct to 2 decimal places. [2]
- b. The diagram shows the graph of the function $y = g(x)$



Draw a graph of $y = \sqrt{g(x)}$, showing any asymptotes and stating its domain and range. [3]

- c. An equation can be expressed in the parametric form

$$\begin{aligned} x &= 2\cos \theta - 1 \\ y &= 2 + 2\sin \theta \end{aligned}$$

Express the equation in Cartesian form. [2]

End of Examination

TRIAL HSC

2020 GHS Extension 1 SOLUTIONS

Section I

MC

1. unit vector = $\frac{\hat{i} - 2\hat{j}}{\sqrt{(1)^2 + (-2)^2}}$

$$= \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$$

D

2. **C**

$$(4\hat{i} - 3\hat{j}) \cdot (3\hat{i} + 4\hat{j})$$

$$= (4 \times 3 - 3 \times 4) = 0 \Rightarrow \text{perpendicular}$$

$$|4\hat{i} - 3\hat{j}| = \sqrt{4^2 + (-3)^2} = 5 \Rightarrow \text{magnitude } 5$$

3. $y = \tan^{-1}(3x)$

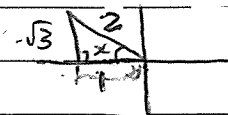
$$\frac{dy}{dx} = \frac{1}{1+(3x)^2} \cdot 3$$

$$= \frac{3}{1+9x^2}$$

C

4. **B**

5. $\sin x = \frac{\sqrt{3}}{2}$ Q2



$$\sin 2x = 2 \sin x \cos x$$

$$= 2 \times \frac{\sqrt{3}}{2} \times \frac{-1}{2}$$

$$= -\frac{\sqrt{3}}{2}$$

B

6. $3! \times 4!$ B

7. $\sin(\alpha + \beta)$; $\sin \alpha \cos \beta + \cos \alpha \sin \beta = a$ — (1)

$\sin(\alpha - \beta)$; $\sin \alpha \cos \beta - \cos \alpha \sin \beta = b$ — (2)

(1) + (2)

$$2 \sin \alpha \cos \beta = a + b$$

$$\sin \alpha \cos \beta = \frac{a+b}{2}$$

D

8. $\frac{y}{2} = \cos^{-1} \left(\frac{x}{2} - 1 \right)$

Domain: $-1 \leq \frac{x}{2} - 1 \leq 1$

$$0 \leq x \leq 4$$

Range: $0 \leq \frac{y}{2} \leq \pi$

$$0 \leq y \leq 2\pi$$

A

9. $\sin 3x \sin 4y = \frac{1}{2} [\cos(3x - 4y) - \cos(3x + 4y)]$

A

(using reference sheet)

10. $P(x) = (x-2)(x+1)Q(x) + Ax + b$

$P(2) = 5 \Rightarrow 2A + B = 5$ — (1)

$P(-1) = 8 \Rightarrow -A + B = 8$ — (2)

(1) - (2)

$$3A = -3$$

$$A = -1$$

substitute $A = -1$ into (1)

$$-2 + B = 5$$

$$B = 7$$

A

(2)

Section II

Question 11 (14 marks)

a) $\frac{3}{4x-1} \geq 2$

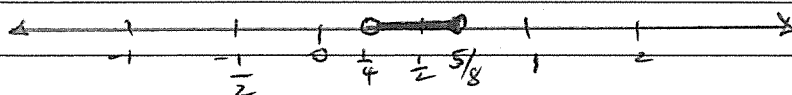
$$x \neq -\frac{1}{4}$$

Solve $\frac{3}{4x-1} = 2$

$$8x - 2 = 3$$

$$8x = 5$$

$$x = \frac{5}{8}$$



Test $x = 0$

$$\frac{3}{4(0)-1} = -3 \leq 2$$

X

Test $x = \frac{1}{2}$

$$\frac{3}{4(\frac{1}{2})-1} = 3 \geq 2$$

✓

Test $x = 1$

$$\frac{3}{4(1)-1} = 1 \leq 2$$

X

$\therefore$ solution : $\frac{1}{4} < x \leq \frac{5}{8}$

b) $y = e^x \tan^{-1} x$

$$\frac{dy}{dx} = vu' + uv'$$

$$= e^x \tan^{-1} x + \frac{e^x}{1+x^2}$$

$$u = e^x, \quad v = \tan^{-1} x$$

$$u' = e^x, \quad v' = \frac{1}{1+x^2}$$

c) i) $f(x) = x^2 - 4x + 5 \Rightarrow$ Axis of symmetry $= \frac{4}{2}$
 $x = 2$

$\therefore$ largest positive domain : $x \geq 2$

ii) c) ii) Points of intersection are on the line $y=x$

i.e. when $x = x^2 - 4x + 5$

$$x^2 - 5x + 5 = 0$$

$$x = \frac{5 \pm \sqrt{25 - 20}}{2}$$

$$= \frac{5 \pm \sqrt{5}}{2}$$

In the restricted domain point of intersection is

$$\left(\frac{5+\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2} \right)$$

iii) For $f(x)$, when $x=2$, $y=1 \Rightarrow$ range is $y \geq 1$

$\therefore$ Domain of inverse function is $x \geq 1$.

iv) $f^{-1}(x)$: $x = y^2 - 4y + 5$

$$x = y^2 - 4y + 4 + 1$$

$$x - 1 = (y - 2)^2$$

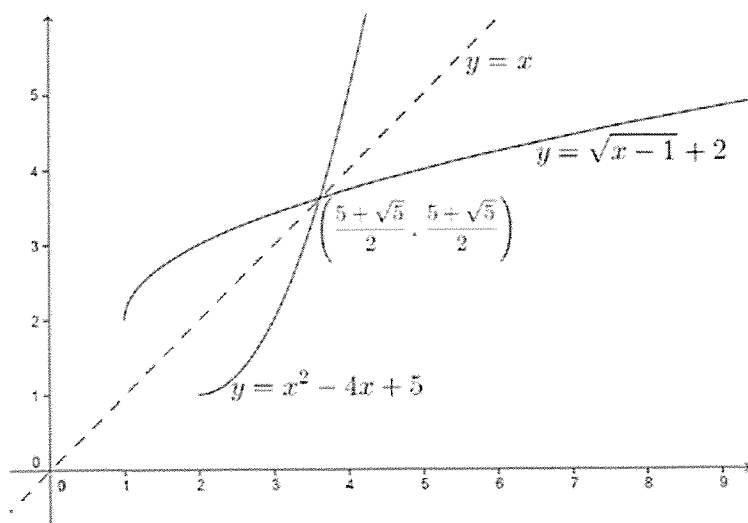
$$y - 2 = \pm \sqrt{x - 1}$$

$$y = 2 \pm \sqrt{x - 1}$$

Since range of inverse is $y \geq 2$

$$y = 2 + \sqrt{x - 1}$$

v)



Question 12 (11 marks)

a) $\int_0^{15} \frac{x}{\sqrt{x+1}} dx$

; $u^2 = x+1$
 $x = u^2 - 1$
 $dx = 2u du$

When $x=0$, $u=1$

$x=15$, $u=4$

$= \int_1^4 \frac{u^2 - 1}{u} \cdot 2u du$

$= 2 \int_1^4 (u^2 - 1) du$

$= 2 \left[\frac{u^3}{3} - u \right]_1^4$

$= 2 \left[\left(\frac{64}{3} - 4 \right) - \left(\frac{1}{3} - 1 \right) \right]$

$= 36$

b) i) $\vec{AO} = \vec{AB} - \vec{OB}$

$\vec{AO} = \vec{OB} - \vec{AB}$ ($\triangle ABO \equiv \triangle OCD$)

$\vec{AB} = \vec{AO} + \vec{OB}$

$= (\vec{v} - \vec{u}) + (\vec{v} - \vec{u})$

$= 2\vec{v} - 2\vec{u}$

ii) $\vec{BD} = \vec{AD} - \vec{AB}$

$= (2\vec{v} - 2\vec{u}) - \vec{v}$

$= \vec{v} - 2\vec{u}$

(5)

$$12 \text{ c) i) } \vec{BA} = \underline{\underline{a}} - \underline{\underline{b}}$$

$$= \underline{\underline{i}} + 3\underline{\underline{j}} - (2\underline{\underline{i}} + \underline{\underline{j}})$$

$$= -\underline{\underline{i}} + 2\underline{\underline{j}}$$

$$\vec{BC} = \underline{\underline{c}} - \underline{\underline{a}}$$

$$= \underline{\underline{i}} - 2\underline{\underline{j}} - (2\underline{\underline{i}} + \underline{\underline{j}})$$

$$= -\underline{\underline{i}} - 3\underline{\underline{j}}$$

$$\text{ii) } |\vec{BA}| = \sqrt{(-1)^2 + 2^2}$$

$$= \sqrt{5}$$

$$|\vec{BC}| = \sqrt{(-1)^2 + (-3)^2}$$

$$= \sqrt{10}$$

$$\text{iii) } \cos \angle ABC = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| \cdot |\vec{BC}|}$$

$$= \frac{(-1 \times -1) + (2 \times -3)}{\sqrt{5} \times \sqrt{10}}$$

$$= \frac{-5}{5\sqrt{2}} = \frac{-1}{\sqrt{2}}$$

$$\angle ABC = 135^\circ$$

Question 13 (13 marks)

a) $3^{2n+1} + 2^{n+2}$ is divisible by 7 for $n \geq 1$

Step 1 Show true for $n=1$

$$\begin{aligned} 3^3 + 2^3 &= 27 + 8 \\ &= 35 \text{ which is divisible by 7} \\ \therefore \text{true for } n=1 \end{aligned}$$

Step 2 : Assume true for $n=k$

$$\begin{aligned} \text{i.e. } 3^{2k+1} + 2^{k+2} &= 7p \quad \text{where } p \text{ is an integer} \\ 3^{2k+1} &= 7p - 2^{k+2} \end{aligned}$$

Step 3 : Prove true for $n=k+1$

$$\text{i.e. } 3^{2k+3} + 2^{k+3} = 7q \quad \text{where } q \text{ is an integer}$$

$$\begin{aligned} \text{LHS} &= 3^{2k+3} + 2^{k+3} \\ &= 3^2 \cdot 3^{2k+1} + 2^{k+3} \end{aligned}$$

$$= 3^2 (7p - 2^{k+2}) + 2^{k+3} \quad (\text{from assumption})$$

$$= 9 \times 7p - 9 \times 2^{k+2} + 2 \times 2^{k+2}$$

$$= 9 \times 7p - 7 \times 2^{k+2}$$

$$= 7(9p - 2^{k+2})$$

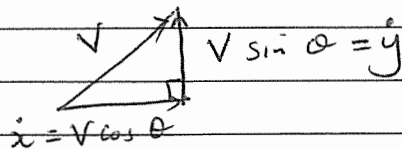
$$\begin{aligned} &= 7q \quad \text{where } q = 9p - 2^{k+2} \\ &= \text{RHS} \end{aligned}$$

$\therefore$ If true for $n=k$ then also true for $n=k+1$

Step 4 : By the principle of Mathematical Induction,
the result is true for all $n \geq 1$

b)

i) Initially



Horizontally

$$\ddot{x} = 0$$

$$\dot{x} = C_1$$

$$\text{when } t=0, \dot{x} = V \cos \theta$$

$$\dot{x} = V \cos \theta$$

$$x = Vt \cos \theta + C_2$$

$$\text{when } t=0, x=0$$

$$\therefore C_2 = 0$$

$$\therefore x = Vt \cos \theta \quad \text{--- (1)}$$

Vertically

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_1$$

$$\text{when } t=0, \dot{y} = V \sin \theta$$

$$\therefore C_1 = V \sin \theta$$

$$\dot{y} = -10t + V \sin \theta$$

$$y = -5t^2 + Vt \sin \theta + C_2$$

$$\text{when } t=0, y=4 \therefore C_2 = 4$$

$$\therefore y = Vt \sin \theta - 5t^2 + 4 \quad \text{--- (2)}$$

ii) From (1)

$$x = Vt \cos \theta \Rightarrow t = \frac{x}{V \cos \theta}$$

substituting into (2)

$$y = V \sin \theta \left(\frac{x}{V \cos \theta} \right) - 5 \left(\frac{x}{V \cos \theta} \right)^2 + 4$$

$$= x \tan \theta - \frac{5x^2}{V^2} \sec^2 \theta + 4$$

$$= x \tan \theta - \frac{5x^2}{V^2} (1 + \tan^2 \theta) + 4$$

$$\left(\tan \theta = \frac{5}{12} \right)$$

$$\text{when } x = 30, y = 0$$

$$\therefore 0 = 30 \times \frac{5}{12} - \frac{5 \times 30^2}{V^2} \left(1 + \left(\frac{5}{12} \right)^2 \right) + 4$$

$$V^2 = 320.076 \dots$$

$$V = 17.87 \text{ m/s (to 2dp)}$$

13 b (iii) max height $\Rightarrow y = 0$

$$-10t + v \sin \theta = 0$$

$$t = \frac{v \sin \theta}{10}$$

$$= \frac{17.89 \times \frac{5}{13}}{10}$$

$$t = 0.688 \text{ s}$$

when $t = 0.688$,

$$y = 17.89 \times 0.688 \times \frac{5}{13} - 5 \times 0.688^2 + 4$$

Max. height = 6.37 m (to 2dp)

c) $P(x) = x^3 + ax^2 + bx + 5$

$$P(1) = 1 + a + b + 5 = 0$$

$$\therefore a + b = -6 \quad \text{--- (1)}$$

$$P'(x) = 3x^2 + 2ax + b$$

$$P'(1) = 3 + 2a + b = 0$$

$$\therefore 2a + b = -3 \quad \text{--- (2)}$$

$$\textcircled{2} - \textcircled{1}$$

$$a = 3$$

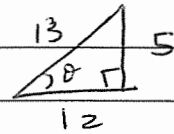
substitute $a = 3$ into $\textcircled{1}$

$$3 + b = -6$$

$$b = -9$$

$$a = 3, \quad b = -9$$

⑨



Question 14 (11 marks)

a) i) $y = x^2$ — (1)

$y = 12 - 2x^2$ — (2)

Solving simultaneously

$$12 - 2x^2 = x^2$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

when $x = 2$, $y = 4$

when $x = -2$, $y = 4$

$\therefore$ Points of intersection $(2, 4)$ and $(-2, 4)$

ii) Rotation about y-axis $\Rightarrow V = \pi \int x^2 dy$

$$V = \pi \int_0^4 y dy + \pi \int_4^{12} (6 - \frac{y}{2}) dy$$

From (1) $x^2 = y$

From (2) $x^2 = \frac{12-y}{2}$

$$= \pi \left[\frac{y^2}{2} \right]_0^4 + \pi \left[6y - \frac{y^2}{4} \right]_4^{12}$$

$$x^2 = 6 - \frac{y}{2}$$

$$= \pi \left[\frac{16}{2} - 0 \right] + \pi \left[6(12) - \frac{12^2}{4} \right] - \left(6(4) - \frac{4^2}{4} \right)$$

$$= 8\pi + 16\pi$$

$$= 24\pi \text{ cubic units.}$$

b) i) $T = 3 + Ae^{-kt}$

$$\Rightarrow Ae^{-kt} = T - 3$$

$$\frac{dT}{dt} = -k \times Ae^{-kt}$$

$$= -k(T - 3)$$

$$\therefore T = 3 + Ae^{-kt} \text{ satisfies the equation}$$

14 b) (ii) $T = 3 + Ae^{-kt}$

when $t=0$, $T=25$

$$25 = 3 + Ae^0$$

$$A = 22$$

$$\therefore T = 3 + 22e^{-kt}$$

when $t=10$, $T=11$

$$\therefore 11 = 3 + 22e^{-10k}$$

$$8 = 22e^{-10k}$$

$$e^{-10k} = \frac{8}{22}$$

$$\log_e e^{-10k} = \log_e \frac{4}{11}$$

$$-10k = \log_e \frac{4}{11}$$

$$k = \frac{-1}{10} \left(\log_e \frac{4}{11} \right)$$

$$= 0.1011 \dots$$

T when $t=15$

$$T = 3 + 22e^{-15(0.1011 \dots)}$$

$$= 7.8241 \dots$$

Temperature after 15 mins $= 7.8^\circ$ (to 1dp)

c) 4 digit numbers $= 2 \times 4 \times 3 \times 2 = 48$

5 digit numbers $= 5!$

Total $= 168$

(11)

Question 15 (11 marks)

c) (i) $2\sin 2x - 3\cos 2x - 3\sin x + 3 = \sin x (6\sin x + 4\cos x - 3)$

$$\begin{aligned} \text{LHS} &= 2\sin 2x - 3\cos 2x - 3\sin x + 3 \\ &= 2(2\sin x \cos x) - 3(1 - 2\sin^2 x) - 3\sin x + 3 \\ &= 4\sin x \cos x - 3 + 6\sin^2 x - 3\sin x + 3 \\ &= 4\sin x \cos x + 6\sin^2 x - 3\sin x \\ &= \sin x (6\sin x + 4\cos x - 3) \\ &= \text{RHS} \end{aligned}$$

ii) $6\sin x + 4\cos x = R\sin(x + \alpha)$
 $= R\sin x \cos \alpha + R\cos x \sin \alpha$

Equating coefficients

$$R\cos \alpha = 6 \quad \text{--- (1)} \quad R\sin \alpha = 4 \quad \text{--- (2)}$$

Squaring and adding (1) & (2)

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 6^2 + 4^2$$

$$R^2 = 52$$

$$R = \sqrt{52}$$

$$R > 0$$

$$\cos \alpha = \frac{6}{\sqrt{52}} \quad ; \quad \sin \alpha = \frac{4}{\sqrt{52}} \quad \therefore \alpha < 90^\circ$$

$$\alpha = 0.588$$

$$\therefore 6\sin x + 4\cos x = \sqrt{52} \sin(x + 0.588)$$

iii) $2\sin 2x - 3\cos 2x - 3\sin x + 3 = 0 \quad 0 \leq x < \pi$

$$\Rightarrow \sin x (6\sin x + 4\cos x - 3) = 0$$

$$\Rightarrow \sin x (\sqrt{52} \sin(x + 0.588) - 3) = 0$$

$$\therefore \sin x = 0 \quad \text{or} \quad \sqrt{52} \sin(x + 0.588) - 3 = 0$$

$$x = 0$$

$$\sin(x + 0.588) = \frac{3}{\sqrt{52}}$$

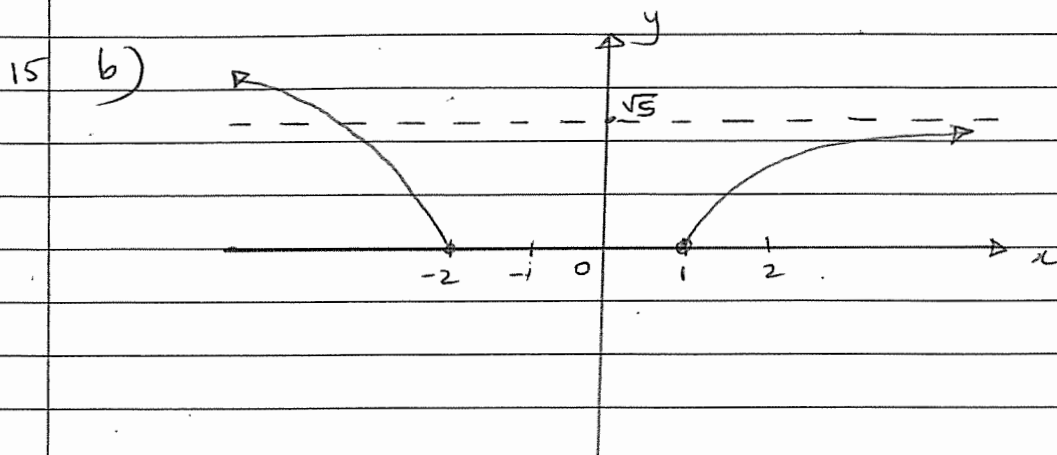
$$x + 0.588 = 0.429 \quad \text{or} \quad 2.7125$$

$$\therefore x = 0 \quad \text{or} \quad x = 2.12$$

$$x = 2.1245 \quad \left(\begin{smallmatrix} \text{since} \\ 0 \leq x < \pi \end{smallmatrix} \right)$$

$$x = 2.12 \quad (\text{to 2 dp})$$

(12)



Domain : $x \leq -2, x \geq 1$

Range : $y \geq 0$

a)

$$x = 2 \cos \theta - 1 \Rightarrow \cos \theta = \frac{x+1}{2}$$

$$y = 2 + 2 \sin \theta$$

$$\Rightarrow \sin \theta = \frac{y-2}{2}$$

Using $\cos^2 \theta + \sin^2 \theta = 1$

$$\left(\frac{x+1}{2} \right)^2 + \left(\frac{y-2}{2} \right)^2 = 1$$

$$\frac{(x+1)^2}{4} + \frac{(y-2)^2}{4} = 1$$

$$(x+1)^2 + (y-2)^2 = 4$$